NBA Player

Performance Analysis

The NBA dataset includes player statistics like points, assists, rebounds, and minutes from the 2013–2014 season. It is used to analyze player performance, team strategies, and the impact of different positions in basketball.

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Why NBA Player Analysis Matters ??

- Helps teams make **better strategies** and game plans

- Identifies **top-performing players** and key contributors

- Assists in **team selection and player recruitment**

- Improves **player performance** by analyzing strengths & weaknesses

- Helps coaches understand **role of each position (PG, SG, SF, PF, C)**

- Useful for **fans and analysts** to understand game trends

Scoring Pattern of Players

- **Score Concentration**

- Most players score between **0 to 500 points**, indicating that the majority have **lower scoring contributions.**

- **High Scorer Presence**

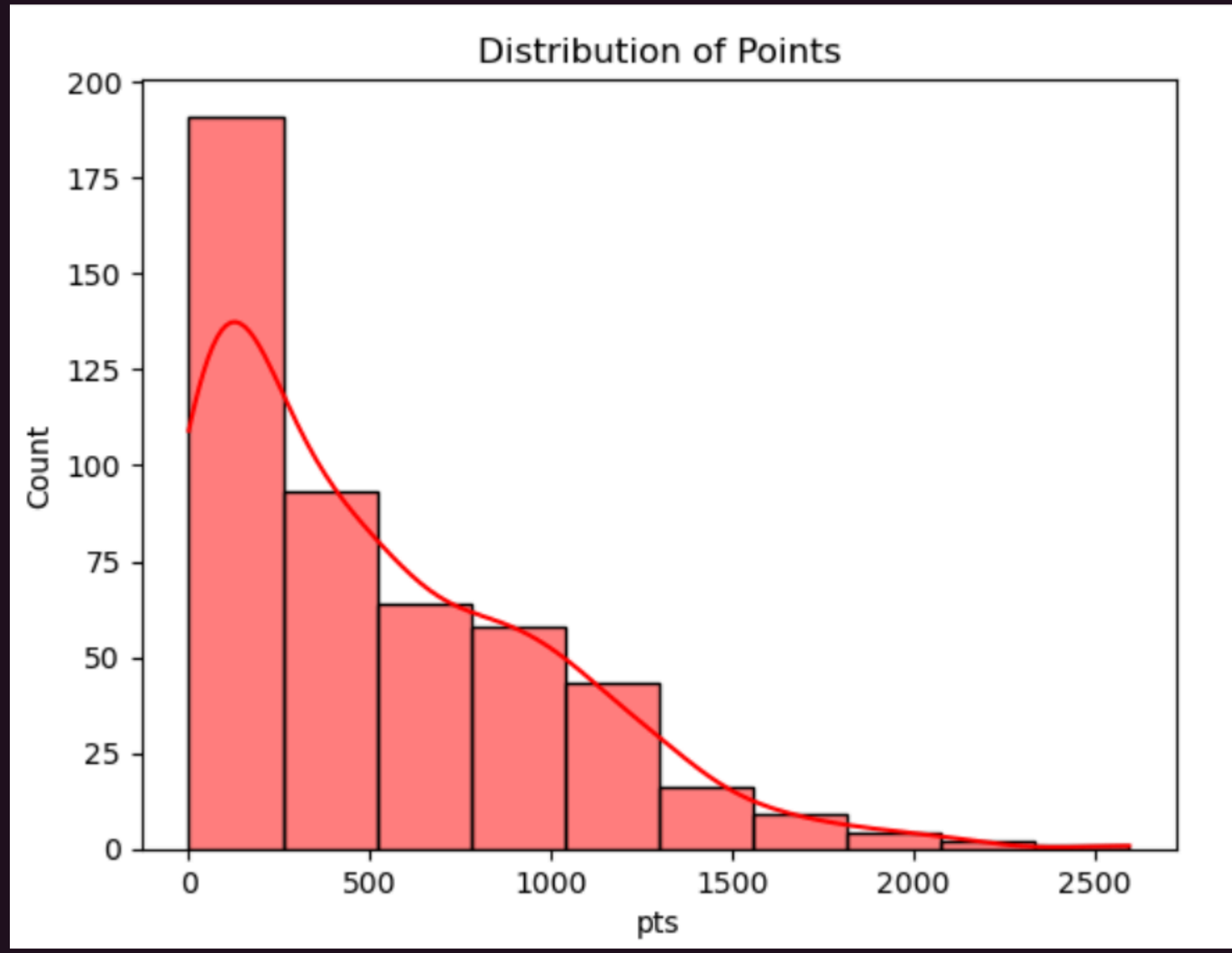
- Only a few players score above **1500 points**, highlighting a small group of **high-performing players.**

- **Distribution Pattern**

- The distribution is **right-skewed**, meaning scoring is **not evenly distributed** among players

- **Performance Imbalance**

- A small number of **star players dominate scoring**, while most players contribute **relatively fewer points.**



Relationship Between Performance Metrics

- **Multi-variable Analysis**

- The pairplot visualizes **pairwise relationships** between key variables such as **points, assists, rebounds, and minutes played.**

- **Points vs Minutes**

- There is a **strong positive relationship** between **points and minutes played,** indicating that players who play more tend to **score more.**

- **Assists Relationship**

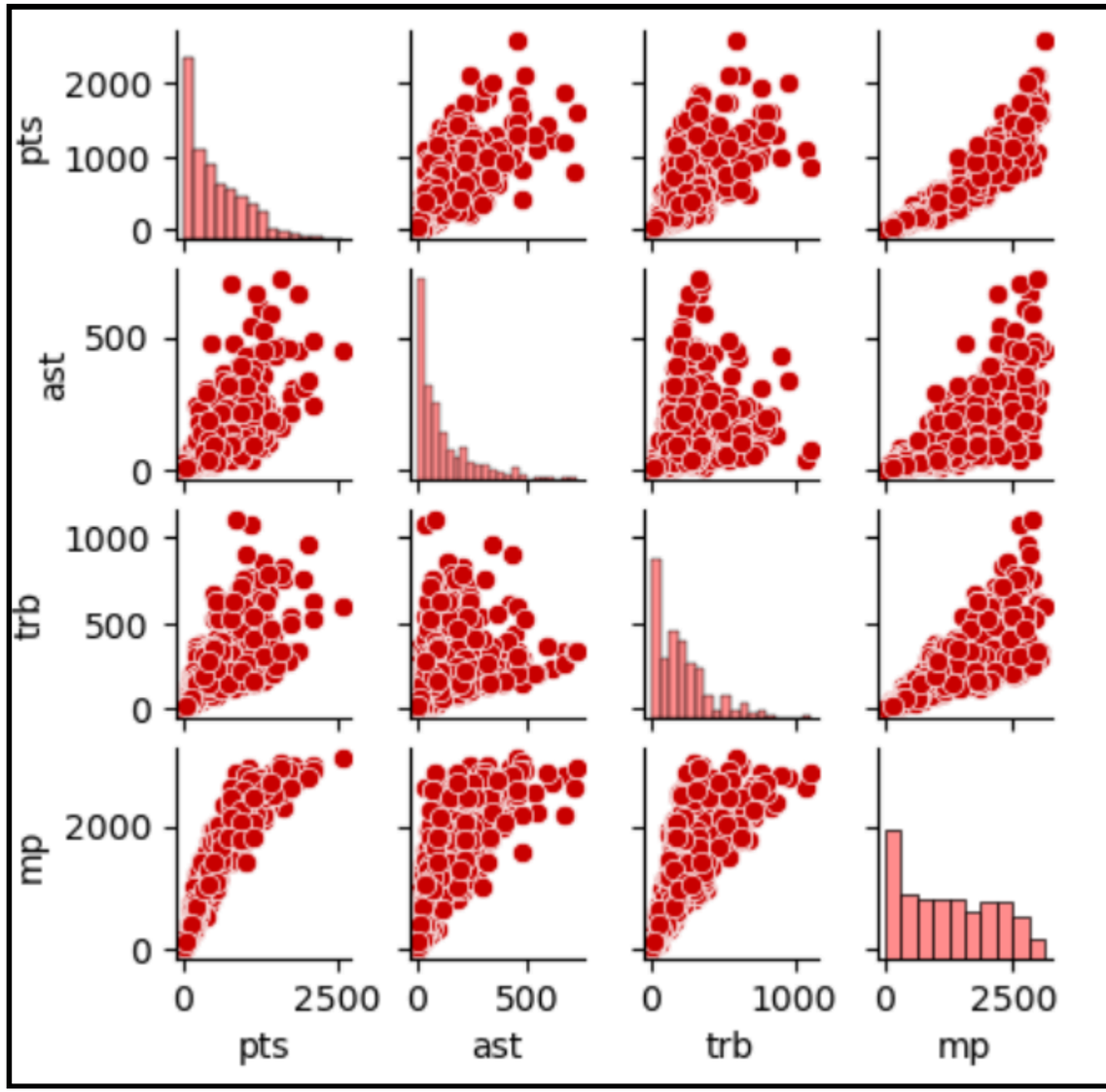
- Assists show a moderate relationship with points, suggesting that playmaking contributes to scoring performance.

- **Rebounds Impact**

- Rebounds do not show a strong relationship with points, indicating they are less directly linked to scoring.

- **Key Performance Driver**

- Overall, **playing time (minutes)** is the **most important factor** influencing scoring performance.



Relationship Between Rebounding and Scoring

Weak Correlation

- There is a **weak relationship** between **rebounds and points,** indicating that rebounding does not strongly influence scoring

Low Performance Cluster

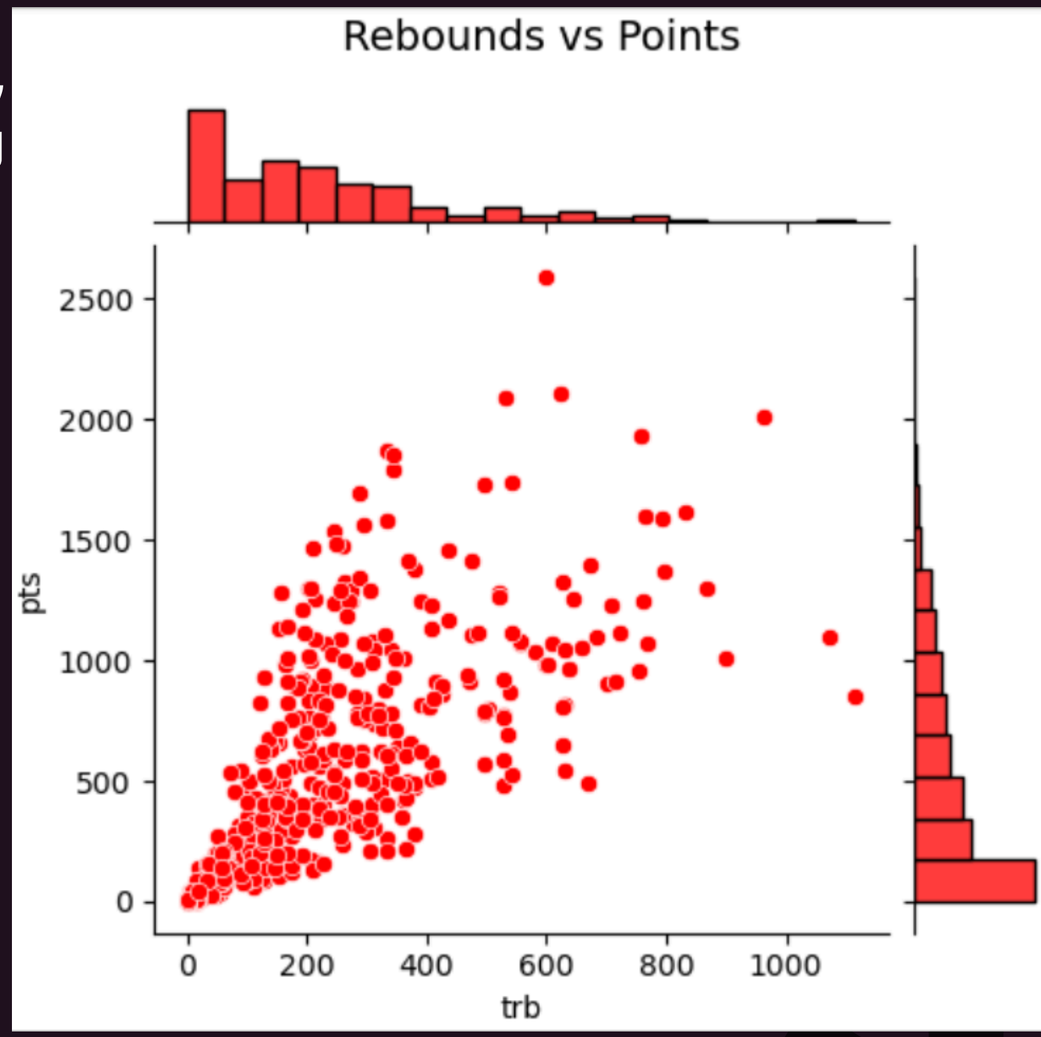
- A large number of players have **low rebounds and low points,** showing a concentration of lower-performing players.

Role Specialization

- Players with **high rebounds do not necessarily score high points,** highlighting **different player roles** in the game.

Overall Insight

- The analysis shows that **rebounding and scoring are not strongly related,** as players specialize in different performance areas.



Positional Contribution to Total Scoring

Highest Scoring Positions

- **Power Forwards (22.1%)** and **Shooting Guards (22.0%)** contribute the **highest share of total points**, making them the primary scorers.

Lowest Contribution

- **Centers contribute only 15.4%**, which is the **lowest among all positions.**

Role Distribution

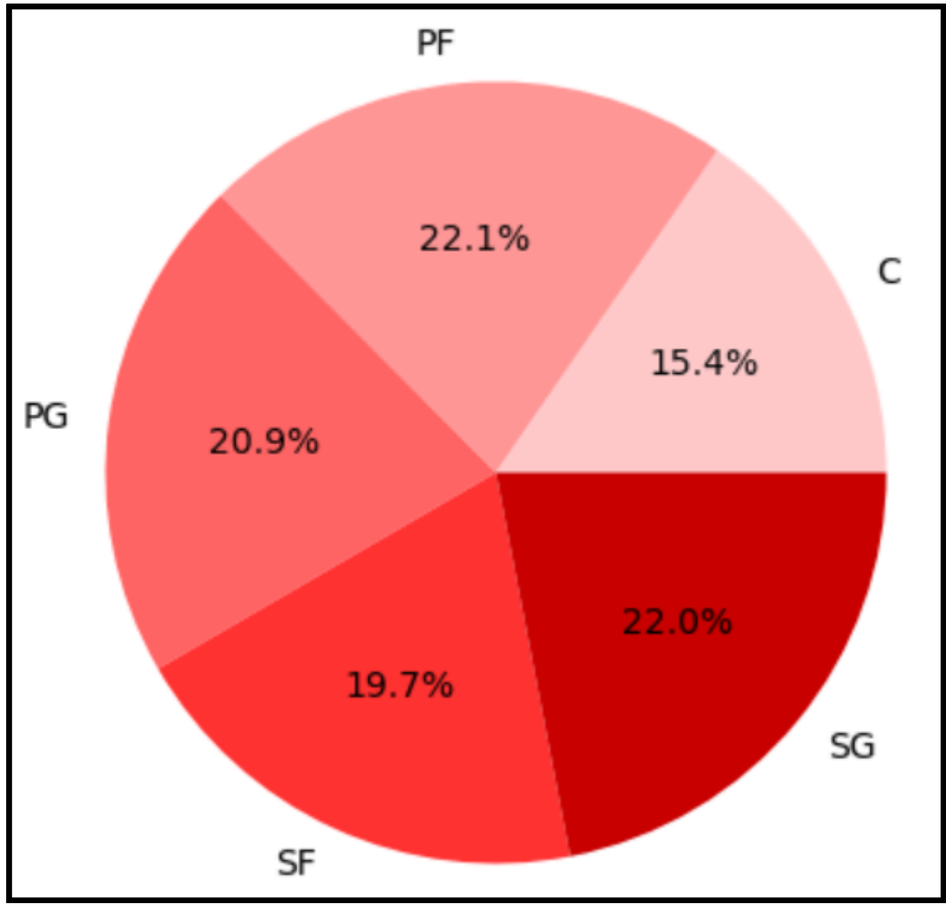
- **Forwards and guards** are the main contributors in scoring, indicating their **offensive importance.**

Supporting Role of Centers

- **Centers play a more supportive role**, focusing less on scoring and more on other aspects of the game.

Overall Insight

- The scoring distribution highlights a clear **role-based contribution**, with guards and forwards leading offensive performance.



Average Scoring Efficiency by Player Position

Highest Scoring Position

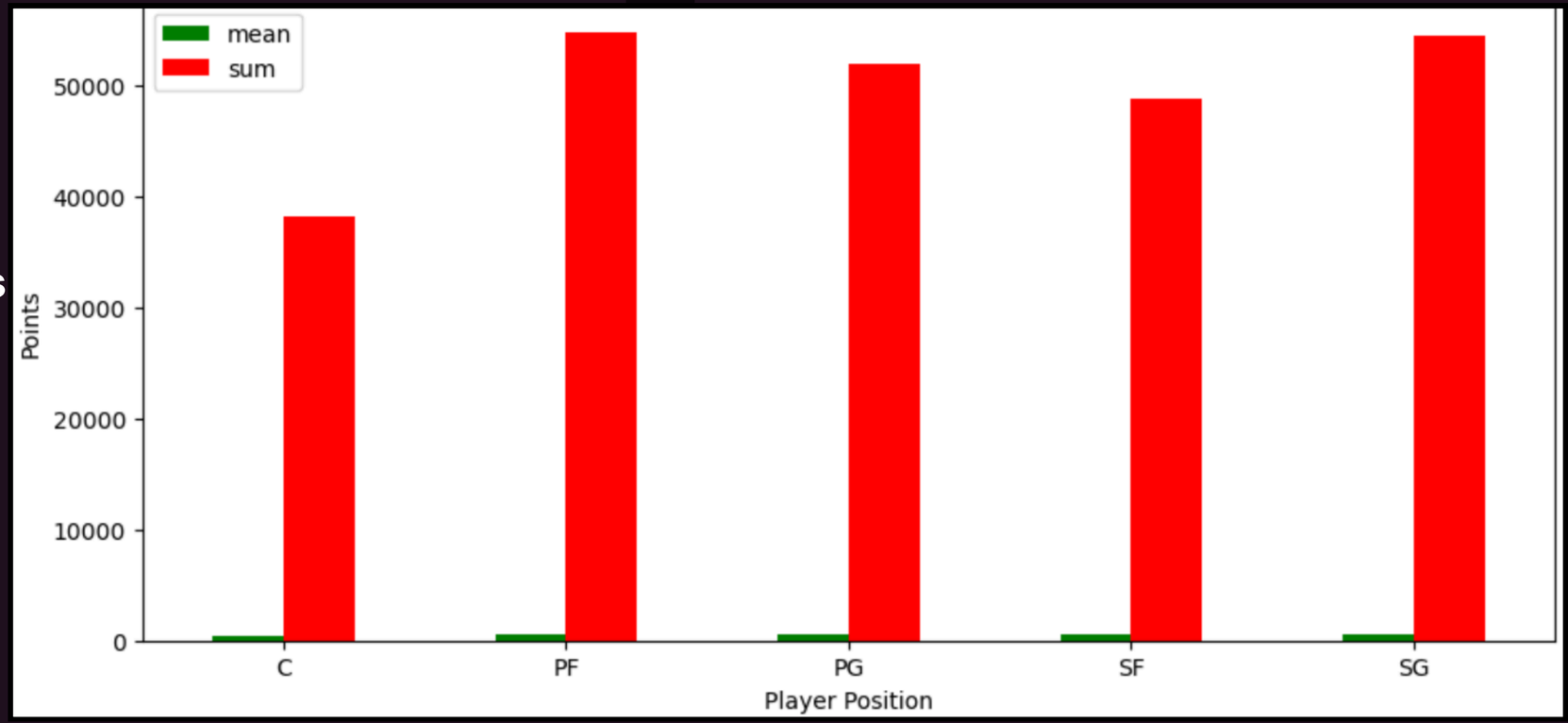
- **Point Guards (PG)** have the highest average points at around **610**, making them the most **efficient scorers.**

Moderate Scoring Positions

- **Power Forwards (PF)** average around **570 points**, followed by **Shooting Guards (SG)** and **Small Forwards (SF)** in the range of **490–500 points.**

Lowest Scoring Position

- **Centers (C)** have the lowest average at approximately **425 points**, indicating lower scoring contribution



Role-Based Performance

- The data shows that **guards (especially PGs)** play a key role in scoring, while centers focus less on scoring and more on other responsibilities.

Overall Insight

- Overall, scoring efficiency is **highest among Point Guards**, highlighting their importance in offensive gameplay.

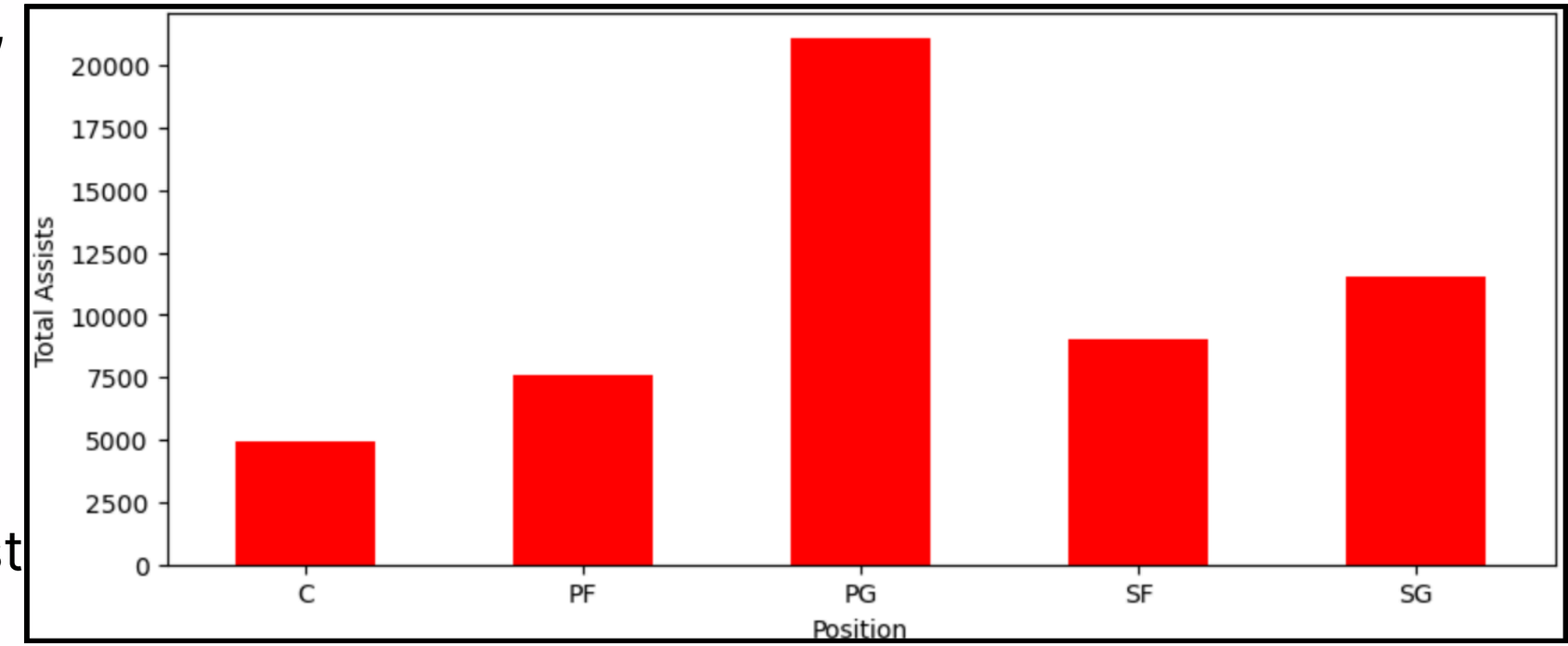
Leading Playmakers Based on Assists

- **Point Guards (PG)** contribute the highest assists (~21,000), showing they are the main **playmakers.**

- **Shooting Guards (SG)** and **Small Forwards (SF)** provide moderate support in assists

- **Centers (C)** have the lowest assists, indicating minimal involvement in playmaking

- Overall, assists are heavily dominated by **guards**, highlighting their role in **game control and passing.**



Evaluating Team Strength and Efficiency in Scoring

team_id	sum
TOT	21502
POR	8753
PHO	8629
DAL	8597
LAC	8582

Values

- POR → **8753**
- PHO → **8629**
- DAL → **8597**
- LAC → **8582**

- These teams have the **highest total points**, showing strong overall team performance

- Scores are **very close (~8500–8700)** → high competition among top teams

- High total indicates **consistent contribution from multiple players**

team_id	mean
GSW	655.833333
TOR	642.100000
DEN	639.833333
IND	612.083333
MIN	609.142857

Values

- GSW → **655.83**
- TOR → **642.10**
- DEN → **639.83**
- IND → **612.08**
- MIN → **609.14**

- GSW has the **highest average points per player**

- Indicates **better efficiency and player performance**

- These teams may not always have highest totals, but have **strong individual contributions**

Team Composition by Player Position (Top 5 Teams)

The top 5 teams have a **higher number of players overall**, indicating stronger squad depth

- Most teams show a **balanced distribution across positions (C, PF, PG, SF, SG)**

- **SF and SG positions appear more frequently**, showing preference for scoring roles

- **Centers (C) are fewer** compared to other positions

- Some teams rely more on specific positions, showing **different team strategies**

- Teams with more players and balanced positions have **better flexibility in gameplay**

- Higher number of **guards and forwards** indicates focus on speed and scoring

- Fewer centers suggest a shift toward **modern, fast-paced basketball**

pos team_id	C	PF	PG	SF	SG
TOT	9	12	10	16	16
NOP	5	3	2	5	3
OKC	3	4	3	3	4
UTA	3	3	4	3	3
LAL	3	3	3	3	4

Final Conclusion

- NBA performance depends on both **team strength and individual efficiency**

- Different positions have **specialized roles** in the game

- Success comes from a **balance of scoring, playmaking, and defense**

Thank You